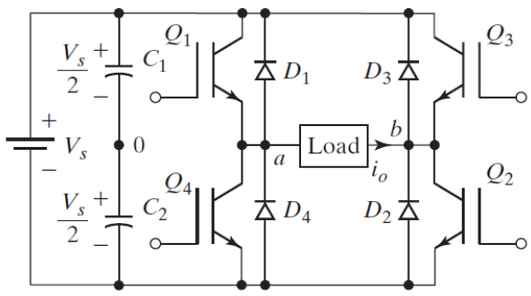
Shahjalal University of Science & Technology
Dept. of Electrical & Electronic Engineering
Course No: EEE 0713 3239
Power Electronics

Assessment #5

Coverage: Week 11

ID: **Time: 30 min** Total points: 20

(1) For the single-phase full-bridge inverter shown in the figure, switches $Q_1$and Q_2conduct from δ to $\pi - \delta$, and switches $Q_3$and Q_4conduct from $\pi + \delta$ to $2\pi - \delta$.
Sketch the output voltage waveform and derive an expression for the RMS value of the output voltage as a function of δ. Also, Calculate the RMS value for $\delta = \pi/6$, assuming a DC input voltage of V_s.



(2) For the problem in Question 1, Determine the Fourier series expression of the output voltage.
(i) Find the RMS value of the fundamental component.
(ii) Calculate the total harmonic distortion (THD) of the output voltage. Evaluate THD for $\delta = \pi/6$